



Coordinate Geometry · Mark Scheme

Chapter 3: Coordinate Geometry. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Straight lines

- 1 $m = \frac{-7-5}{4-(-2)} = -2$ M1 A1; $y-5 = -2(x+2) \Rightarrow y = -2x+1$ A1.
- 2 $y = \frac{3}{4}x+2$: gradient $\frac{3}{4}$ B1; crosses y-axis at (0, 2) B1.
- 3 Gradient of $2x+5y=7$ is $-\frac{2}{5}$ B1; $y+1 = -\frac{2}{5}(x-3)$ M1; $2x+5y-1=0$ A1.
- 4 Perpendicular gradient $-\frac{1}{3}$ B1; $y-4 = -\frac{1}{3}(x+1)$ M1; $x+3y-11=0$ A1; $y=0 \Rightarrow (11, 0)$ B1 ft.
- 5(a) $m_{AB} = \frac{7-3}{4+2} = \frac{2}{3}$ M1; $m_{BC} = \frac{4-7}{6-4} = -\frac{3}{2}$ A1; $\frac{2}{3} \times (-\frac{3}{2}) = -1$, so $AB \perp BC$ A1 – conclusion needed.
- 5(b) $AB = \sqrt{6^2+4^2} = \sqrt{52}$, $BC = \sqrt{2^2+3^2} = \sqrt{13}$ M1 A1; area = $\frac{1}{2}\sqrt{52}\sqrt{13} = \frac{1}{2}\sqrt{676} = 13$ A1.
- 5(c) $D = A + C - B$ (the vector $\vec{BC}$ applied to A) M1; $D = (0, 0)$ A1.

Section B – Circles

- 6 $(x-3)^2 + (y+5)^2 = 16$ B1 B1 (LHS, RHS).
- 7 Centre $(-2, 7)$ B1; $r = \sqrt{45} = 3\sqrt{5}$ B1.
- 8 $(x+4)^2 - 16 + (y-5)^2 - 25 + 5 = 0$ M1 A1; $(x+4)^2 + (y-5)^2 = 36$; centre $(-4, 5)$ A1; radius 6 A1.
- 9 Centre = midpoint = $(3, 5)$ B1; $PQ = \sqrt{8^2+6^2} = 10$ so $r = 5$ M1 A1; $(x-3)^2 + (y-5)^2 = 25$ A1.
- 10(a) $(x-3)^2 + (y+1)^2 = 10-k$ M1; centre $(3, -1)$ A1.
- 10(b) Need $r^2 > 0$: $10-k > 0$ M1; $k < 10$ A1.
- 10(c) Touches y-axis, so radius = distance from centre to y-axis = 3 M1 A1; $10-k = 9 \Rightarrow k = 1$ A1.

Section C – Circle geometry

- 11 $(4-1)^2 + (1+3)^2 = 9+16$ M1 = 25, so P lies on C A1.
 - 12 Centre $(1, -3)$; gradient of radius = $\frac{1-(-3)}{4-1} = \frac{4}{3}$ M1 A1; tangent gradient $-\frac{3}{4}$ M1; $y-1 = -\frac{3}{4}(x-4) \Rightarrow 3x+4y-16=0$ A1.
 - 13 $m_{CA} = \frac{6-2}{1+3} = 1$, $m_{CB} = \frac{6-2}{1-5} = -1$ M1 A1; product = -1 , so $\angle ACB = 90^\circ$; by the converse of the angle-in-a-semicircle theorem, C lies on the circle A1.
 - 14(a) $x^2 + (x+1)^2 = 13$ M1; $x^2 + x - 6 = 0 \Rightarrow x = 2, -3$ A1; $A(2, 3), B(-3, -2)$ A1.
 - 14(b) $AB = \sqrt{5^2+5^2}$ M1 = $5\sqrt{2}$ A1.
 - 15(a) $(x-2)^2 + (y-1)^2 = 20$ M1 A1; centre $(2, 1)$, radius $2\sqrt{5}$ A1.
 - 15(b) Substitute: $(x-2)^2 + (2x+k-1)^2 = 20$ M1; $x^2 - 4x + 4 + 4x^2 + 4(k-1)x + (k-1)^2 - 20 = 0$ M1; $5x^2 + (4k-8)x + k^2 - 2k - 15 = 0$ (given answer) A1.
 - 15(c) Tangent $\Rightarrow b^2 - 4ac = 0$: $(4k-8)^2 - 20(k^2 - 2k - 15) = 0$ M1; $k^2 + 6k - 91 = 0$ A1; $(k+13)(k-7) = 0 \Rightarrow k = 7$ or $k = -13$ A1.
- Alternative: perpendicular distance from centre to the line equals the radius, $\frac{|3+k|}{\sqrt{5}} = 2\sqrt{5}$.



Section D – Parametric to Cartesian

16 $t = x - 2$ M1; $y = (x - 2)^2$ A1.

17 $t = \frac{x}{3}$ M1; $y = \frac{18}{x}$ (or $xy = 18$) A1.

18 $\cos \theta = \frac{x}{4}$, $\sin \theta = \frac{y}{4}$; use $\sin^2 \theta + \cos^2 \theta = 1$ M1; $x^2 + y^2 = 16$ A1; circle, centre O , radius 4 B1.

19 $\cos \theta = \frac{x-2}{3}$, $\sin \theta = \frac{y+1}{3}$ M1; $\left(\frac{x-2}{3}\right)^2 + \left(\frac{y+1}{3}\right)^2 = 1$ M1; $(x-2)^2 + (y+1)^2 = 9$, a circle A1; centre $(2, -1)$, radius 3 A1.

20(a) $t = \frac{x-3}{4}$ M1; $y = (x-3) + 8 + \frac{5}{2 \cdot \frac{x-3}{4}} = x + 5 + \frac{10}{x-3}$ M1 A1; $= \frac{(x+5)(x-3) + 10}{x-3} = \frac{x^2 + 2x - 5}{x-3}$ A1; $t > 0 \Rightarrow x > 3$ B1.

20(b) $4t + 3 = 5 \Rightarrow t = \frac{1}{2}$ M1; check $y = 2 + 8 + 5 = 15$ A1.

Section E – Parametric sketching and points on a curve

21(a) $x: 4, 1, 0, 1, 4$; $y: -4, -2, 0, 2, 4$ B1 B1. **21(b)** Parabola opening to the right, vertex at O , arrow pointing upward (from $(4, -4)$ to $(4, 4)$) B1.

22 $t^2 - 4 = 0 \Rightarrow t = \pm 2$ M1 A1; $(3, 0)$ and $(-1, 0)$ A1.

23 (a) $2t + 3 = 9 \Rightarrow t = 3$; check $x = 9 - 1 = 8$ M1 A1. (b) $t = -3$: $x = 8$, $y = -3$ B1.

24 $t^2 - 3 = 2t + 5$ M1; $t^2 - 2t - 8 = 0$ A1; $t = 4$ or $t = -2$ M1 A1; $A(8, 13)$, $B(-4, 1)$ A1.

25(a) $y = 0$: $t(t^2 - 4) = 0 \Rightarrow t = 0, \pm 2$ M1 A1; points $(-4, 0)$ and $(0, 0)$ A1.

25(b) $x = 0 \Rightarrow t = \pm 2$ M1; then $y = 0$ in both cases, so only the origin A1.

25(c) $y = t \cdot x$ M1; so $y^2 = t^2 x^2$ and $t^2 = x + 4$ M1; $y^2 = x^2(x + 4)$ A1.

Teaching point: the curve passes through the origin twice (at $t = \pm 2$) – a loop, just like the worked example in the notes.

Roar-level challenge

26 Radius $= \sqrt{3^2 + 4^2} = 5$ M1; $(x-3)^2 + (y-4)^2 = 25$ A1; gradient of radius $\frac{4}{3}$, so tangent gradient $-\frac{3}{4}$ M1; $3x + 4y = 0$ A1.